FANR 3800: Week 1 Student Notes

Spatial Analysis for Natural Resources · Fall 2026

Invalid Date

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1 Week 1 - Everything Has a Spatial Component?

Name: ____ **Date:** __________________

1.1 How to Use This Sheet: Two-Pass Notes

In class, capture the key idea under each prompt in your own words: a phrase or a sentence, not everything. That evening, reread your notes and use the "After class: add detail" box at the bottom of each page to add detail from the assigned ESRI Training exercise and the eLC links. Need more room? Use your own paper.

(Week 1 note: we have no previous lab yet, so a couple of sections are adapted for our first week.)

1.2 1 · This Week *(10 min)*

Weekly schedule, ESRI Training, deliverables, and due dates.

1.3 2 · Course & GIS Orientation *(10 min)*

In one sentence, what is GIS as a way of thinking, not just software?

A management question from my discipline that is really a "where?" question.

1.4 3 · The Week Ahead *(5 min)*

New ideas this week: *Spatial Thinking*, the *Vector Data Model*, the *Raster Data Model*.

1.5 4 · Application Discussion *(10 min) (may vary; instructor's call)*

Read the prompt for **your** focus and jot a response; be ready to share.

- **Forestry/Forest Management:** As a new forestry student, how do you envision GIS being used to manage timberlands in the pine forests of the Southeast?
- **Wildlife Management:** Why is a consistent file-naming convention important for a biologist tracking endangered-species sightings in the Everglades?
- **Fisheries Management:** As a fisheries biologist studying a large reservoir, what "spatial data" might you collect, and how could GIS help you understand it?
- **Parks Recreation & Tourism Management:** For a manager at a Georgia Piedmont state park, how could GIS help manage visitor access or trail maintenance?

1.6 5 · Lecture: What GIS Is and Why Your Problems Are Spatial *(15 min)*

- GIS is data, software, methods, hardware, and people, used to ask and answer spatial questions.

- Why every natural resource decision has a spatial component.

- The five arcs (A to E): the one-line story of each.

Key terms: GIS · spatial data · arc

1.7 After Class: Add Detail

From the assigned ESRI Training exercise and the eLC links, add one concrete example of GIS used in *your* discipline, and a definition of "spatial data" in your own words.

2 WEDNESDAY

2.1 1 · Lecture: How We Represent the World, the Two Data Models *(20 min)*

- *Vector Data Model*: points, lines, polygons, each with an attribute table. One natural-resource example of each.

- *Raster Data Model*: a grid of cells, one value each; *Spatial Resolution* and cell size are related; integer vs. floating-point vs. multiband.

- How do I decide whether a dataset should be vector or raster?

Key terms: vector · raster · cell / resolution · attribute table

2.2 2 · Getting Ready for Lab: Why It Matters *(10 min)*

What will getting set up (ESRI account and ArcGIS Pro) let me do, and why do it now?

2.3 3 · Getting Ready for Lab: Steps & Tools *(10 min)*

The setup steps and what I will meet in the interface: ESRI account, ArcGIS Pro access, training account; the project, map view, and Catalog.

2.4 4 · Week Summary *(10 min)*

In your own words, the one thing to remember from Week 1:

Questions I still have (bring to Monday):

Deliverables: ESRI account and ArcGIS Pro access by the start of Week 2; first ESRI Training module before next Wednesday.

2.5 After Class: Add Detail

> From the assigned ESRI Training exercise and the eLC links, add one dataset you would store as vector and one you would store as raster in your field, and note why.

2.6 Evening Review Checklist (Second Pass)

- ☐ Reread my in-class notes within 24 hours
- ☐ Added detail from the assigned ESRI Training exercise
- ☐ Added detail from the eLC web links
- ☐ Wrote my own one-sentence summary of the week
- ☐ Listed questions to ask on Monday